

Task 1 Report

Task 1: Data Cleaning and Preprocessing

Introduction

Data preprocessing is an essential step in any data science project. The objective of this task was to clean and prepare the e-commerce sales dataset for further analysis and machine learning applications.

Dataset Description

The dataset used in this project contains e-commerce transaction records including:

- Order ID
- Customer ID
- Product ID
- Product Category
- Price
- Discount
- Quantity
- Payment Method
- Order Date
- Delivery Time
- Region
- Return Status
- Total Amount
- Shipping Cost
- Profit Margin
- Customer Age
- Customer Gender

Data Cleaning Process

Dataset Loading

The dataset was imported into Python using the Pandas library and examined to understand its structure.

Missing Value Analysis

Missing values were identified using the `isnull().sum()` function. Numerical columns were treated using median imputation, while categorical columns were filled using the most frequent value (mode).

Duplicate Record Handling

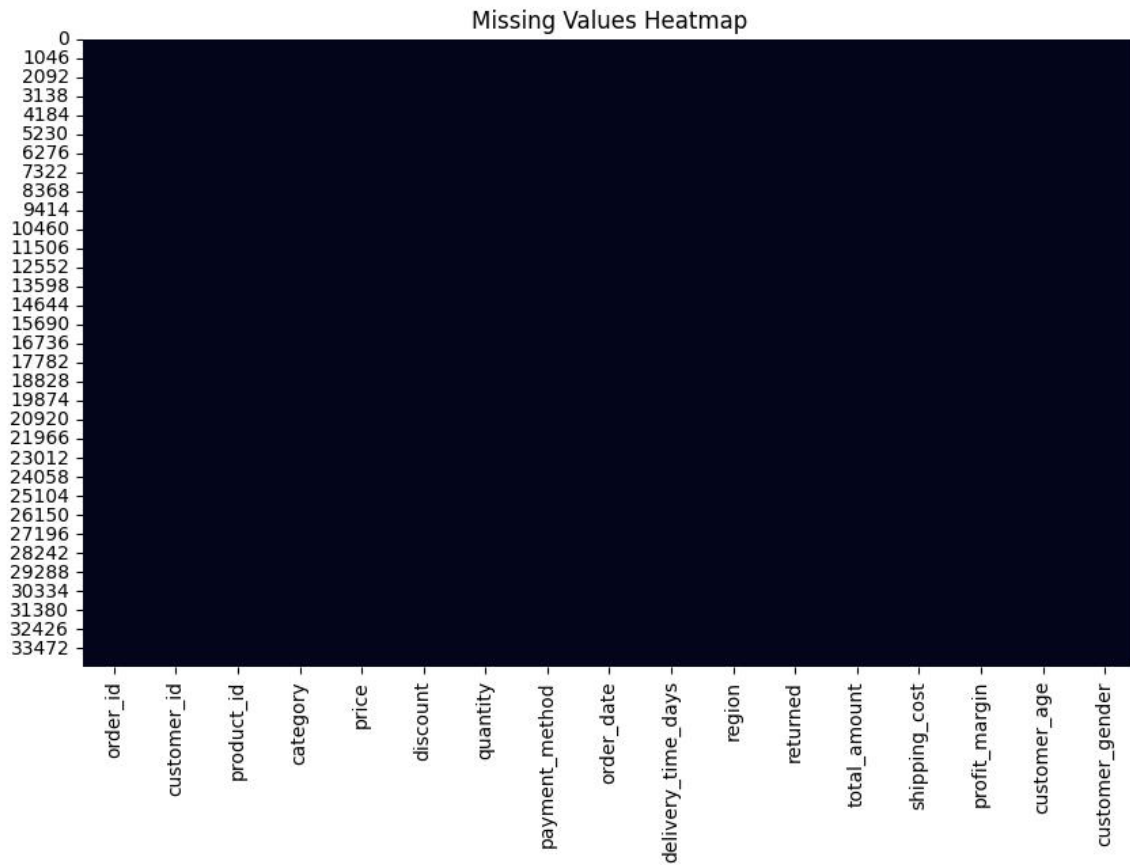
Duplicate records were checked using the `duplicated()` function and removed where necessary to ensure data consistency.

Data Type Conversion

The `order_date` column was converted into datetime format to support future date-based analysis.

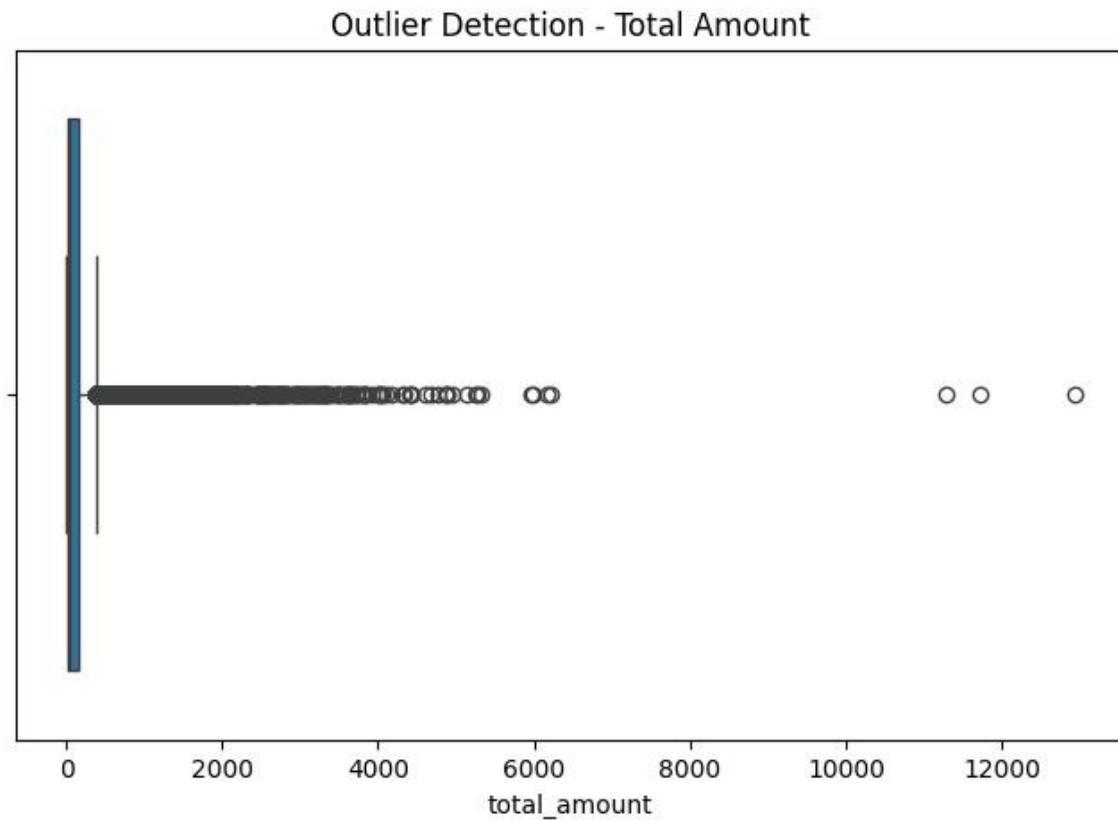
Missing Value Visualization

A heatmap was generated using Seaborn to visually inspect missing values across the dataset.



Outlier Detection

Outliers in the total_amount column were analyzed using boxplots and the Interquartile Range (IQR) method.



Cleaned Dataset Export

The processed dataset was saved as `cleaned_ecommerce_sales.csv` for use in subsequent tasks.

Results

- Missing values successfully handled.
- Duplicate records removed.
- Date column converted to proper format.
- Outliers identified and analyzed.
- Cleaned dataset exported successfully.

Conclusion

The dataset was successfully cleaned and standardized. The resulting dataset is suitable for exploratory data analysis and predictive modeling tasks.